

## ORIGINAL RESEARCH

## CORONARY, PERIPHERAL, AND STRUCTURAL INTERVENTIONS

# A Machine Learning Driven Approach to Quantifying Coronary Artery Tortuosity



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## ABSTRACT

**BACKGROUND** The associations between coronary artery tortuosity and age, sex, and cardiovascular risk factors are not fully established. Prior studies enrolled fewer than 1,000 subjects and relied on heuristic metrics.

**OBJECTIVES** The purpose of this study was to develop and validate an automated right coronary artery (RCA) tortuosity measure using machine learning (ML)-based vessel segmentation and examine its association with sex, age, coronary artery disease (CAD), and cardiovascular risk factors.

**METHODS** We developed an ML-enabled pipeline to quantify RCA tortuosity as a continuous measure in 38,691 RCA angiograms from 22,334 patients and evaluated concordance with blinded interventional cardiologist review in a subset of 300 angiograms. Regression models were used to study the association of tortuosity with age, sex, hypertension, diabetes, hypercholesterolemia, smoking, and presence of CAD.

**RESULTS** Tortuosity ranged from 0.007 to 0.289. Blinded interventional cardiologist classification of high tortuosity vs not had 85.6% precision. Tortuosity was higher in women ( $\beta$  per SD = 0.17;  $P < 0.001$ ) and higher with hypertension ( $\beta$  per SD = 0.06;  $P = 0.002$ ), but lower with diabetes ( $\beta$  per SD = -0.11,  $P < 0.001$ ). After adjustment for risk factors, age was not independently associated with tortuosity. Hypercholesterolemia and smoking were not associated. Higher tortuosity was associated with CAD (OR per SD = 1.05;  $P < 0.001$ ), severe CAD (OR per SD = 1.09;  $P < 0.001$ ), and higher Gensini score ( $\beta$  per SD = 0.05;  $P < 0.001$ ), even after adjustment.

**CONCLUSIONS** We derived a scalable ML-enabled measure of RCA tortuosity from coronary angiography and found associations with sex, hypertension, diabetes, and presence and severity of CAD. (JACC Adv. 2026;5:102829) © 2026 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS  
AND ACRONYMS****CAD** = coronary artery disease**LAO** = left anterior oblique**LDL-C** = low-density  
lipoprotein cholesterol**ML** = machine learning**RCA** = right coronary artery**T2DM** = type 2 diabetes  
mellitus

Coronary artery disease (CAD) remains the leading cause of mortality and morbidity in the world,<sup>1</sup> yet understanding of the genetic and clinical drivers of disease remains limited. Prior research efforts have relied on: 1) large data sets with limited phenotypes—such as genetic association studies using a binary diagnosis code for CAD;<sup>2,3</sup> and 2) small data sets with deep phenotyping, including coronary physiology, intravascular imaging, or biomarkers.<sup>4</sup>

Coronary angiography is the gold standard diagnostic test for CAD, as it provides anatomic characterization of coronary atherosclerosis, which can prognosticate disease and guide treatment decisions.<sup>5</sup> Coronary angiography films are a rich source of information about the patient and disease state, including coronary tree anatomy, plaque distribution and burden, and degree of calcification, but remain underutilized in the research setting. The adoption of machine learning (ML) methods in health care has the potential to change how these data are used. One key feature in coronary imaging is arterial tortuosity, defined as high curvature or excessive bending or kinking of the coronary vessel structure.<sup>6</sup> Visual and automated computational approaches have been proposed to quantify coronary tortuosity, but the field has not arrived at a consensus on a rigorous, standardized tortuosity metric.<sup>6</sup>

Prior studies performed tortuosity measurements manually based on binary visual coding (eg, tortuous vs nontortuous) and pattern descriptions (eg, S- or C-shaped elongations, kinking, looping, coiling).<sup>7-9</sup> While useful for initial phenotyping, they lack continuous measures and suffer from poor reproducibility and interobserver noise.<sup>10</sup> Earlier tortuosity metrics, such as the Tortuosity Index,<sup>7</sup> Sum-of-Angles,<sup>11</sup> and Inflection-Count Metric,<sup>12</sup> provide reproducible scalar values, yet their reliance on global quantities (arc length, total turning angle, inflection count) leads to oversight of local morphological cues. As a result, a curve with more twists and turns can be assigned the same score as a smoother one.<sup>13</sup>

Within the field of cardiology, computer vision techniques have enabled the extraction of imaging features at unprecedented scale from echocardiograms<sup>14</sup> and cardiac magnetic resonance imaging.<sup>15</sup> Studies have shown that multiple ML-derived biomarkers can be derived with computer vision (eg, aortic size, body fat distribution, cardiac fibrosis, liver fat)<sup>15-17</sup> and used to reveal associations with cardiovascular and metabolic diseases, improving

our understanding of cardiac disease endophenotypes and subclinical imaging features. However, large-scale, standardized computations of coronary arterial tortuosity have not yet been performed, nor has the relationship between tortuosity, CAD, and related cardiovascular risk factors been explored.<sup>6</sup>

Preliminary studies showed that coronary tortuosity is associated with lower risk of CAD,<sup>18,19</sup> but positively correlated with risk factors such as hypertension<sup>20</sup> and conditions such as spontaneous coronary artery dissection.<sup>21</sup> Some research also suggests that tortuous arteries are more common in older patients and females.<sup>22-25</sup> However, these studies have been limited by small sample sizes and heuristic measurements of tortuosity. Previous image processing of coronary angiography has mainly focused on segmenting coronary arteries for comparison with quantitative coronary angiography rather than on extracting clinically relevant phenotypes.<sup>26-28</sup> Here, we develop a scalable and reproducible ML-enabled approach to measure coronary artery tortuosity, apply it to measure tortuosity in 38,691 right coronary artery (RCA) angiograms from 22,334 patients, and describe the associations between tortuosity and age, sex, cardiovascular risk factors, and CAD.

**METHODS**

**STUDY COHORT.** This study was conducted using data from the Massachusetts General Hospital cardiac catheterization registry. This registry includes 81,798 coronary angiograms performed between 2000 and 2021 on 31,513 patients with mapping of deidentified Digital Imaging and Communications in Medicine (DICOM) images and metadata. From this registry, we derived a novel ML-powered approach to quantify RCA tortuosity on 38,691 right coronary angiography loops from 22,334 patients using the left anterior oblique (LAO) view. This study used an LAO view with primary angle ranging from  $-10^{\circ}$  to  $100^{\circ}$ , wider than the conventional range of  $-5^{\circ}$  to  $95^{\circ}$ ,<sup>29</sup> to account for measurement variability and ensure comprehensive inclusion of LAO images.<sup>30</sup> Patients without an RCA angiogram loop in LAO view ( $n = 5,172$ ), with poor angiography quality or artifacts such as wires and devices over the coronary ( $n = 1,438$ ), and with inappropriate length of the RCA related to magnification or panning ( $n = 2,569$ ) were excluded (Supplemental Figure 1). In this context, “38,691 angiograms” denotes angiographic loops rather than distinct procedures or follow-up angiograms; temporal follow-up information was not used in this analysis. Some patients contributed more than

one eligible LAO RCA loop from the same procedure ( $n = 10,534$ ; median 2 loops per patient among those with more than 1 eligible loop), reflecting multiple acquisitions during a single angiogram study. Tortuosity from all eligible loops was used for training the transformer model, whereas patient-level analyses used one loop per individual, selecting the loop in which the RCA appeared longest. We did not exclude patients with prior coronary revascularization 28.9% ( $n = 6,459$ ) had a history of percutaneous coronary intervention and 15.5% ( $n = 3,455$ ) had a history of coronary artery bypass grafting. The resulting cohort demographics and characteristics are shown in **Table 1**. Analyses conducted under this study were approved by the Massachusetts General Hospital Institutional Review Board (protocol number 2018P002292).

## SEGMENTATION OF THE RIGHT CORONARY ARTERY

We employed ML techniques to quantify RCA tortuosity from the raw input of the angiography of the RCA in the LAO view in a fully automated manner, eliminating the need for human annotation. Computing tortuosity requires a centerline representation of the artery. To extract this representation, segmentation masks of the RCA were first obtained and then skeletonized from the background of the image. Below, we detail the steps in this process.

First, we trained a U-Net convolutional neural network<sup>31</sup> to segment the RCA on an independent external data set, ARCADE<sup>28</sup> (**Supplemental Methods**). The ARCADE angiogram data set comprises 1,500 labeled frames (one representative frame per angiographic loop) with ground-truth segmentation masks and SYNTAX score annotations.<sup>32</sup> We used a subset of 1,000 images for training and selected the threshold that maximized the Dice coefficient over the remaining 500 images held out for validation. The Dice coefficient<sup>33,34</sup> is defined as:

$$\text{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}$$

where  $P$  and  $G$  denote the predicted and ground-truth masks, respectively. By quantifying their overlap, the Dice coefficient yields a score between 0—indicating no shared region—and 1—indicating perfect spatial agreement—providing a clear measure of segmentation accuracy<sup>35</sup> (**Supplemental Table 1**).

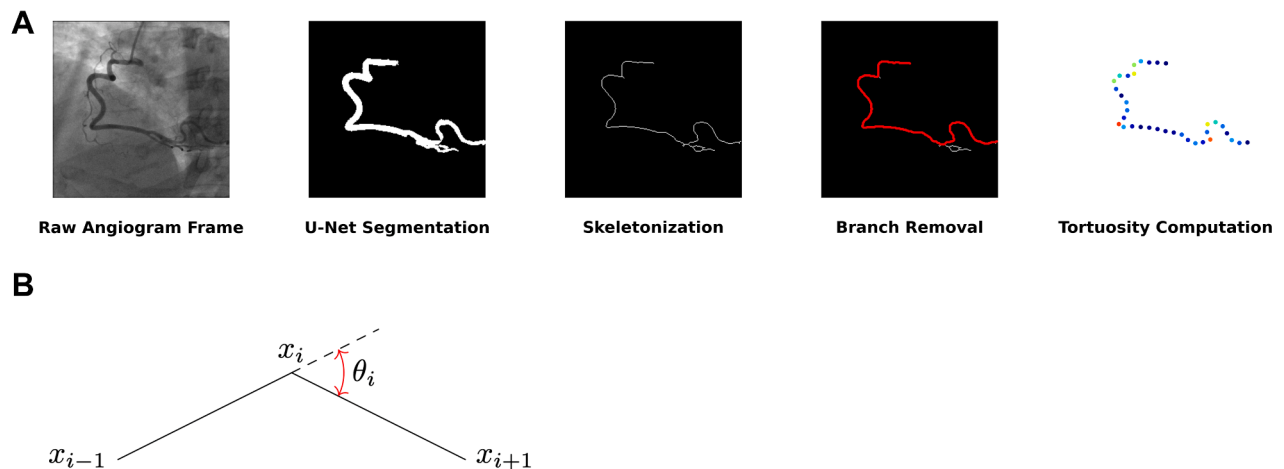
**TABLE 1** Baseline Characteristics of the Study Population

|                             | Total<br>(N = 22,334) | Female<br>(n = 8,265, 37.0%) | Male<br>(n = 14,069, 63.0%) |
|-----------------------------|-----------------------|------------------------------|-----------------------------|
| Mean age (SD), range y      | 64.7 (12.5) 30–102    | 67.6 (12.5) 30–100           | 63.0 (12.3) 30–102          |
| Race                        |                       |                              |                             |
| White, n (%)                | 18,699 (83.7%)        | 6,877 (83.2%)                | 11,822 (84.0%)              |
| Asian, n (%)                | 626 (2.8%)            | 204 (2.5%)                   | 422 (3.0%)                  |
| Black, n (%)                | 834 (3.7%)            | 374 (4.5%)                   | 460 (3.3%)                  |
| Other, n (%)                | 2,175 (9.7%)          | 810 (9.8%)                   | 1,365 (9.7%)                |
| Hypertension, n (%)         | 19,092 (85.5%)        | 7,139 (86.4%)                | 11,953 (85.0%)              |
| Type 2 diabetes, n (%)      | 7,233 (32.39%)        | 2,731 (33.0%)                | 4,567 (32.5%)               |
| Hypercholesterolemia, n (%) | 18,196 (81.5%)        | 6,491 (78.5%)                | 11,705 (83.2%)              |
| Smoking, n (%)              | 8,456 (38%)           | 2,730 (33%)                  | 5,726 (41%)                 |
| Stenosis in RCA, n (%)      | 10,748 (48.1%)        | 3,299 (39.9%)                | 7,449 (52.9%)               |

RCA = right coronary artery.

Next, we applied the trained U-Net segmentation model to our cohort. We segmented the middle 20 frames of each coronary angiography loop to coincide with peak diastolic epicardial flow. This is when the coronary arteries are most visible, as flow predominates during diastole and motion artifacts are minimized.<sup>36</sup> We then skeletonized the segmented masks to obtain a center line representation of the RCA. As the arterial branches define a nonregular curve (ie, the branching pattern defines multiple local trajectories that emerge from a single position, and curvature must be defined on a single centerline), we selected the main branch as the longest path from the beginning of the RCA to the distal end of the posterior descending artery. The main-branch selection was performed automatically, with no manual intervention. This approach maximizes the captured length of the artery while minimizing disconnected segments and reducing artifacts.

From each set of 20 frames per angiographic loop, we automatically identified the frame with the longest visible artery to represent the artery for that loop because it was most likely to accurately capture the vessel's geometry. We retained loops in which that artery had at least 20 or more discrete points identified along the arterial wall with no interruption, which ensured that we had selected a relatively complete artery rather than artifacts (eg, the catheter or a short vessel segment due to blockage). This final filtering process resulted in a total of 38,691 angiograms from 22,334 patients with a high-fidelity segmented RCA ready for tortuosity computation (**Supplemental Figure 1**).

**FIGURE 1** Right Coronary Artery Tortuosity Computation Using Artificial Intelligence

(A) Study design, extraction of right coronary artery tortuosity. (B) Local tortuosity computation. From the raw angiogram frame we obtained its segmentation mask through U-Net Model, from the segmented mask the centerline is extracted, we remove the branches, and then we proceed to compute the tortuosity. Panel (B) shows how local tortuosity is computed: 3 consecutive centerline points  $x_{i-1}, x_i, x_{i+1}$  define adjacent segments; the turning angle  $\theta_i$  at  $x_i$  quantifies local tortuosity.

**QUANTIFICATION OF TORTUOSITY.** The skeletonization of the segmented artery gives us a discrete collection of points that trace out the underlying curve. We compute the *discrete curvature* associated with these points<sup>37</sup>:

**Discrete curvature.** Let  $x_{i-1}$ ,  $x_i$ , and  $x_{i+1}$  be 3 consecutive extracted points along the skeleton of the artery (ie, forming 2 consecutive edges). The *discrete curvature* at  $x_i$  is the local change in direction at  $x_i$ , defined as the turning angle required to align edge  $x_{i-1}x_i$  with edge  $x_i x_{i+1}$  (Figure 1B).

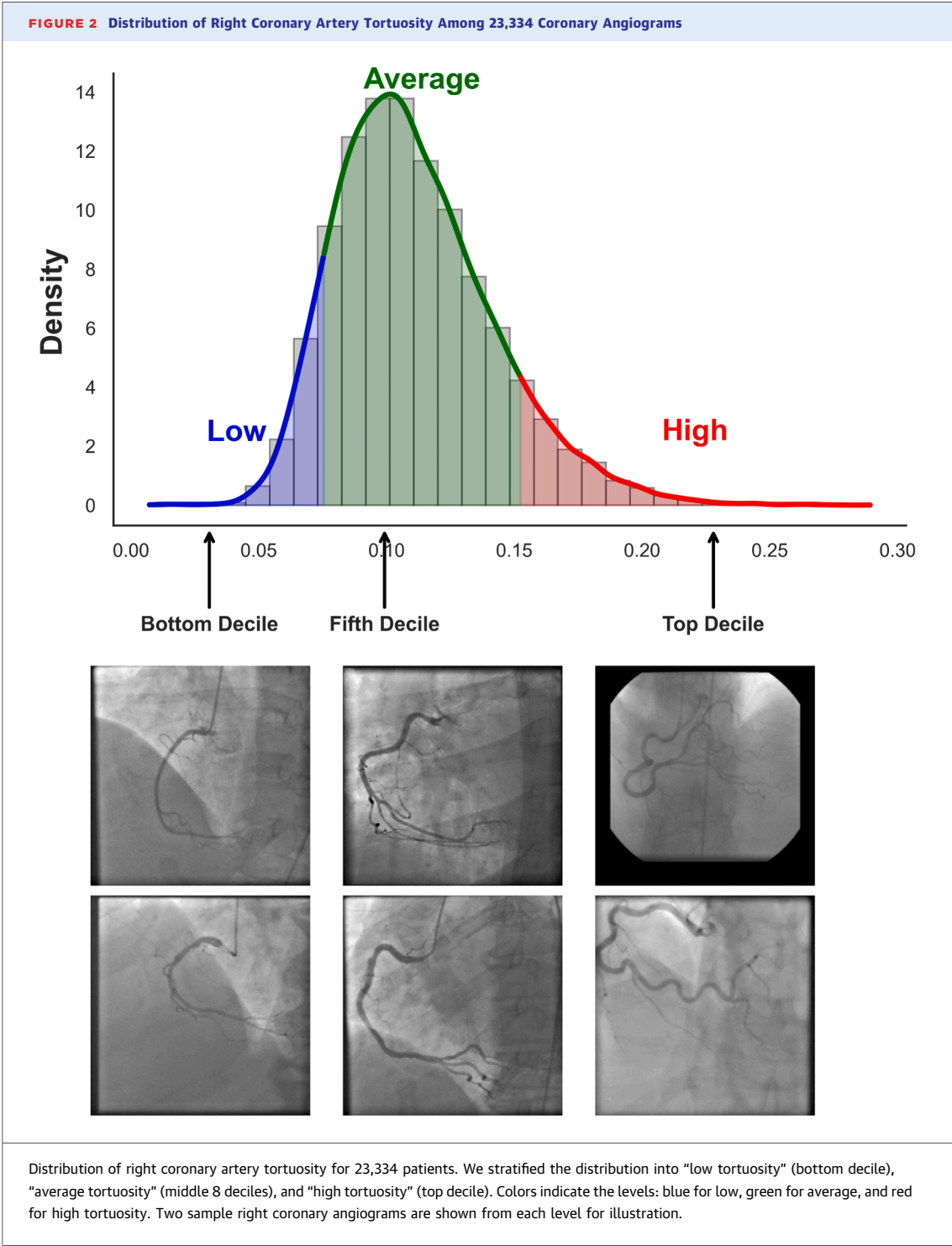
This definition has been previously used by Bribiesca<sup>38</sup> to study the tortuosity of retinopathy fundus images but has not been applied to at scale analysis of arterial coronary tortuosity.<sup>38</sup> Unlike prior methods for computing curvature, the discrete-curvature approach is invariant to translation, rotation, and scaling,<sup>38,39</sup> which is essential for a standardized measurement of tortuosity.<sup>6,10</sup> We average the absolute magnitudes of discrete curvature angles at each skeletonized point and can then be averaged to produce a single global tortuosity measure.

We take the absolute value of each intersegment angle,  $|\theta_i|$ , because the sign of  $\theta_i$  depends only on the traversal direction ( $x_{i-1} \rightarrow x_i$  vs  $x_i \rightarrow x_{i-1}$ ) and conveys no morphological information; the angle's magnitude

is invariant to this choice. Finally, we normalize this measure by  $\pi$  (the maximum turning angle in radians) to produce a tortuosity metric that has a continuous score between 0 (lowest tortuosity) and 1 (and highest tortuosity).

In our results, we validate our score by first stratifying it into low tortuosity (bottom decile), average tortuosity (middle 8 deciles), and high tortuosity (top decile) (Figure 2). We then randomly sampled 100 angiograms from each of the 3 categories (total 300 angiograms). For concordance testing against expert review, the 3 strata were dichotomized into non-tortuous (low and average categories;  $n = 200$ ) and tortuous (high category;  $n = 100$ ). Because the high category was the smaller subgroup, the power calculation was based on this subset, using previously published pairwise cardiologist-cardiologist interobserver variability in coronary angiography as the reference benchmark (22.6%),<sup>40</sup> 100 cases provided 82.53% power at a one-sided  $\alpha$  of 0.05 to distinguish this expected discordance from an unacceptable discordance rate of 35%.

A board-certified interventional cardiologist, blinded to the computed tortuosity values, decile thresholds, and ML-assigned labels, then classified each angiogram as tortuous or nontortuous based on visual assessment, and we compared this expert



classification to our ML-assigned classification. Angiograms were presented in a randomly shuffled order to avoid ordering bias.

**CORONARY ARTERY DISEASE AND RISK FACTORS.** The patient’s age and cardiovascular risk factors were obtained from electronic health records at the time of catheterization. The presence of CAD was obtained from the findings associated with the angiogram from which we computed the tortuosity. CAD was defined as any presence of stenosis (mild or more) in the RCA

(primary analysis) or anywhere in the coronary tree (sensitivity analysis), as determined by the interventional cardiologist performing the coronary angiography. A continuous measure of angiographic burden of CAD was obtained using the modified Gensini score.<sup>41</sup>

To determine the presence of the remaining risk factors—hypertension, type 2 diabetes, smoking, and hypercholesterolemia—we used a combination of diagnostic codes, lab measurements, and usage of outpatient medications. International Classification of Diseases (ICD) codes were extracted from the start of the patient's record to 3 months after the angiogram. Smoking status was determined using ICD codes, consistent with prior published work.<sup>42</sup> Systolic blood pressure and diastolic blood pressure, glycated hemoglobin, low-density lipoprotein cholesterol (LDL-C) levels, and the use of antihypertensive and lipid-lowering medications were extracted from the most recent outpatient labs and prescription records up to 3 months after the catheterization and were used to study the effect change.

From these extracted measurements, we estimated untreated values of LDL-C by adjusting for intake of lipid-lowering medications by applying a constant-multiplier correction (multiplying the observed LDL-C by  $1/(1-0.30) \approx 1.43$ ).<sup>43,44</sup> The estimated untreated values of systolic blood pressure and diastolic blood pressure were obtained by adjusting for intake of antihypertensive medications by adding 10 mm Hg to systolic and 5 mm Hg to diastolic values.<sup>45,46</sup> Finally, based on these values, we determined the presence of the risk factors using the logic in [Supplemental Table 2](#).

**STATISTICAL ANALYSES.** We examined the association between our global tortuosity score and: 1) age and sex; 2) 4 cardiovascular risk factors (hypertension, hypercholesterolemia, type 2 diabetes, and smoking; and 3) the presence of CAD in the RCA or entire coronary tree and its burden by Gensini score using multilinear regression models. The Python language with the *statsmodels* statistical computing library version 0.13.5 was used for analyses.

**TRANSFORMER-BASED LOCAL TORTUOSITY MODELING FOR CAD CLASSIFICATION.** We also tested whether the collection of individual tortuosity angles might provide a stronger association with CAD than our global tortuosity score. To this end, we designed a local-attention neural network transformer that operates on the ordered vector of local tortuosity along the artery, enabling the model to learn which vessel segments contribute most to CAD

discrimination, together with age and sex ([Supplemental Methods](#)). We trained the transformer to predict CAD given age, sex, and the local tortuosity vector. We compared the C-statistic of this locally aware tortuosity model in classifying CAD in the RCA compared to logistic regression model using the global tortuosity score.

## RESULTS

**DISTRIBUTION OF RIGHT CORONARY TORTUOSITY IN MORE THAN 38,000 ANGIOGRAMS.** Our ML-derived coronary tortuosity score was calculated in 38,681 coronary angiograms from 22,334 patients (mean age  $64.7 \pm 12.5$  years, 37% female, 83.7% White) ([Table 1](#)). RCA tortuosity ranged from 0.007 (lowest tortuosity) to 0.289 (highest tortuosity), with a median (IQR) of 0.107 (0.029-0.190). The distribution of tortuosity had a skewness of 0.744 and kurtosis of 3.879, indicating a heavier right tail than a standard normal distribution (the kurtosis of a standard normal is 3). Extreme values above the 99th percentile ( $>0.199$ ;  $n = 224$ ) are visible in the right tail but represent just 1.00% of the cohort ([Figure 2](#)). The 90th percentile threshold for high tortuosity in the overall cohort was 0.152 ([Supplemental Figure 2](#), [Supplemental Table 3](#)). This threshold was similar across race strata (range: 0.151-0.153) and modestly higher in women than in men (0.158 vs 0.148) ([Supplemental Figure 2](#), [Supplemental Table 3](#)). Sex-stratified adjusted linear regression models showed similar covariate patterns in women and men, supporting that the main associations are not driven by a single subgroup ([Supplemental Figure 3](#)). The decile cutoffs were used for stratification and visualization, but the tortuosity score was analyzed as a continuous measure in the association models. As an absolute reference point, the top-decile cutoff clustered around  $\sim 0.15$  in this cohort ([Figure 2](#), [Supplemental Figure 2](#), [Supplemental Table 3](#)), which corresponds to roughly  $\sim 27^\circ$  of average angulation. On an NVIDIA T4 GPU (16 GB), tortuosity calculation required  $1.476 \pm 0.576$  seconds per 20-frame video loop (mean  $\pm$  SD;  $n = 1,000$ ).

**CONCORDANCE BETWEEN CLINICIAN RATINGS AND ML-DERIVED TORTUOSITY SPECTRUM.** A total of 300 coronary angiograms were analyzed manually for inspection. Based on the computed tortuosity metric, the images were distributed into 2 categories: 200 were classified as “nontortuous” and 100 as “tortuous.” The board-certified cardiologist's evaluation yielded agreement rates of Performance metrics for the 2 confidence buckets are summarized in

**Supplemental Table 4.** In the low bucket ( $n = 200$ ), the algorithm achieved precision = 85.6%, recall = 83.5%, and F1 = 84.6%. In the high bucket ( $n = 100$ ), it achieved precision = 75.3%, recall = 58.0%, and F1 = 65.5%. Overall agreement beyond chance was moderate, with Cohen's  $\kappa = 0.52$  (95% CI: 0.41-0.63).

#### ASSOCIATION BETWEEN TORTUOSITY AND CAD.

Each SD increase in our measure of tortuosity was associated with a 5% increase in the odds of presence of CAD. In a multivariable logistic regression adjusting for age, sex, hypertension, type 2 diabetes mellitus (T2DM), hypercholesterolemia, and smoking, this corresponded to an OR of 1.05 (95% CI: [1.03-1.08]  $P < 0.001$ ) (Supplemental Figure 4, Supplemental Table 5).

We next assessed the relationship between tortuosity and CAD severity—categorized as *mild* (5-49% stenosis), *moderate* (50-69% stenosis), or *severe* (70-100% stenosis)—using a multinomial logistic regression model, with the absence of CAD as the reference category (Supplemental Table 6 and Supplemental Figure 5). Among CAD severity categories, higher tortuosity was associated with severe CAD (OR: 1.09; 95% CI: 1.06-1.13;  $P < 0.001$ ), indicating that higher tortuosity is associated with increased odds of severe disease. Tortuosity was not significantly associated with mild or moderate CAD ( $P = 0.154$  and  $P = 0.956$ , respectively). Additionally, both age and sex were consistently positive and significant across all severity levels, suggesting that older patients and those who are male have higher odds of CAD severity.

To study the relationship between tortuosity and burden of disease, we fit linear regression models with the Gensini score<sup>40</sup> as the dependent variable (Supplemental Figure 6, Supplemental Table 7). When considering burden of disease as the Gensini score (not adjusting for collaterals), an increase in SD of tortuosity has an increase of 0.05 (95% CI: 0.04-0.07;  $P < 0.001$ ) per SD of Gensini score.

#### ASSOCIATION OF TORTUOSITY WITH AGE AND SEX.

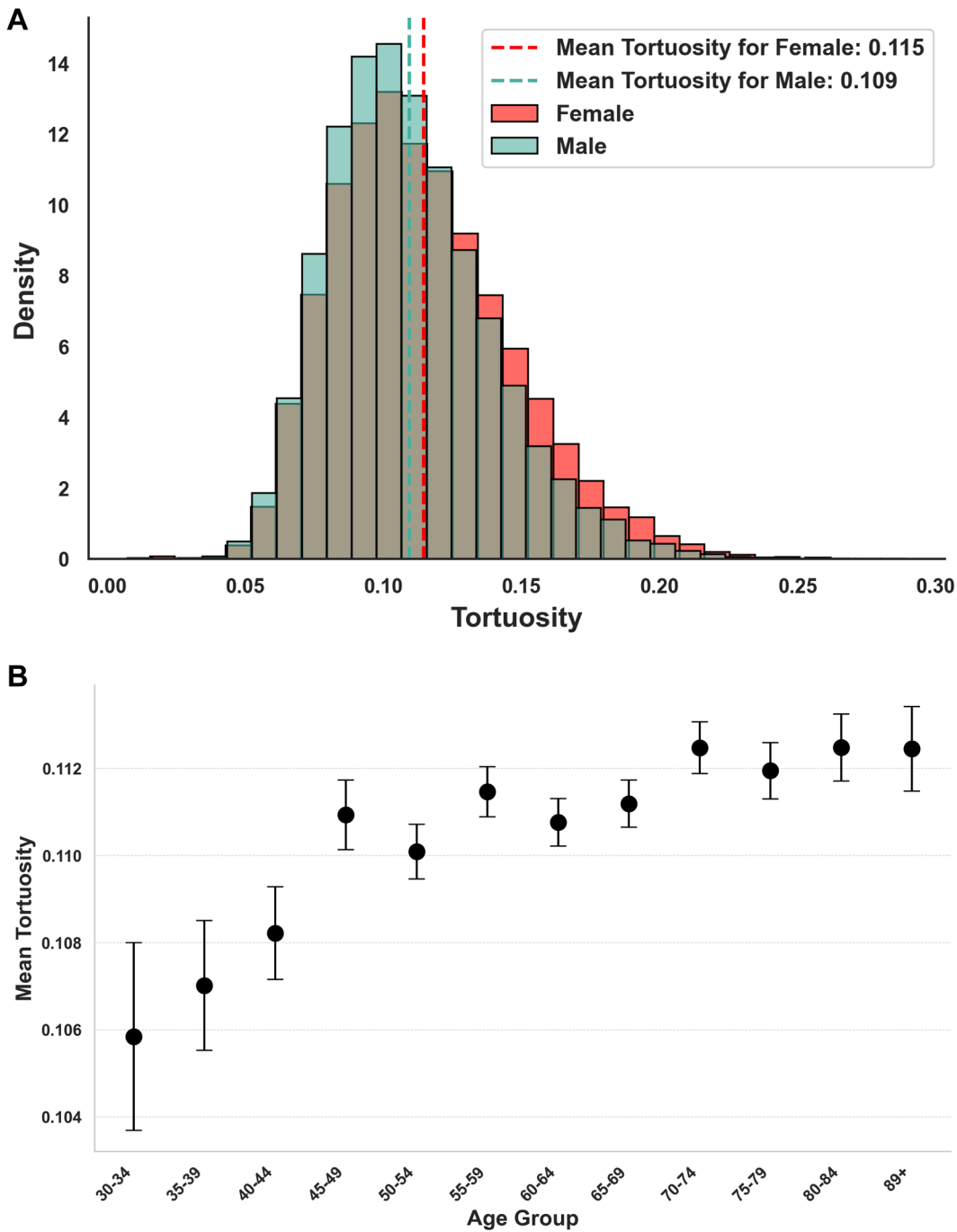
Women exhibited higher tortuosity compared to men (Figure 3A). Age predicted tortuosity in an age- and sex-adjusted model but lost significance after further adjustment for risk factors. To study the relationship between tortuosity and demographic factors, we fit 2 linear regression models with tortuosity as the dependent variable. The first model adjusted for age and sex as clinical predictors. The second adjusted for age, sex, hypertension, T2DM, hypercholesterolemia, and smoking. In a base model including age and sex, age was a significant predictor

of tortuosity,  $\beta = 0.02$ ; 95% CI: 0.01-0.03 per SD of age. However, when cardiovascular comorbidities were added to the model, the age effect was attenuated (adjusted  $\beta = 0.01$ ; 95% CI: -0.00 to 0.03) and lost statistical significance ( $P = 0.121$ ), suggesting that the age-tortuosity association may be mediated by underlying cardiovascular disease. Moreover, using men as the reference group, female sex was associated with higher tortuosity ( $\beta = 0.17$ ; 95% CI: 0.14-0.20;  $P < 0.001$ ), indicating that women exhibited more tortuous arteries than men (Supplemental Table 8). When stratified by 5-year age groups, women had higher mean tortuosity than men across most age categories, with the separation becoming more apparent from midlife onward (Supplemental Figure 7). We also tested an age and sex interaction to assess whether the age-related change in tortuosity differed by sex. Women appeared to have a steeper age-related increase in tortuosity than men, supported by the age and sex interaction term ( $\beta = 0.04$  per SD of age; 95% CI: 0.02-0.07;  $P = 0.002$ ) (Supplemental Table 9).

#### ASSOCIATION BETWEEN TORTUOSITY AND RISK FACTORS.

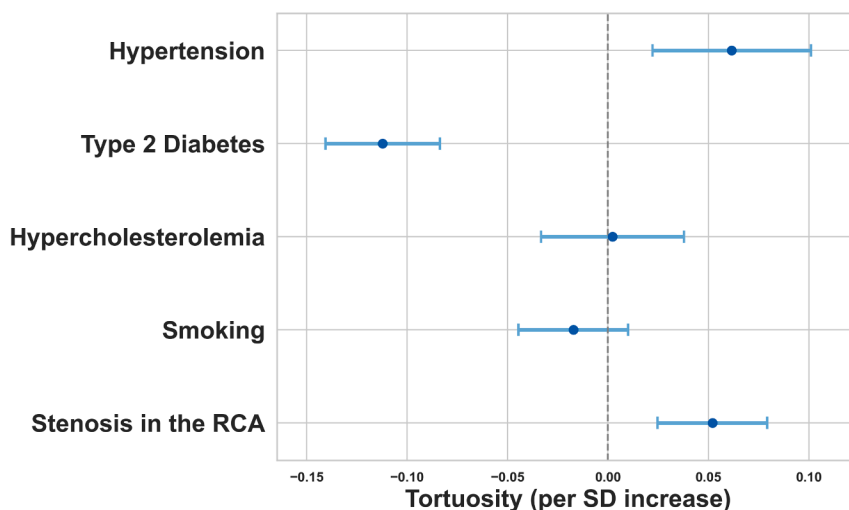
With a multivariable linear regression models predicting tortuosity—adjusted for age, sex, hypertension, T2DM, hypercholesterolemia, smoking status, and presence of stenosis in the RCA—hypertension was positively associated with tortuosity ( $\beta = 0.06$ ; 95% CI: 0.02-0.10;  $P = 0.002$ ), whereas T2DM was inversely associated ( $\beta = -0.11$ ; 95% CI: -0.14 to -0.08;  $P < 0.001$ ). Hypercholesterolemia did not show a significant effect ( $\beta = -0.00$ ; 95% CI: -0.03 to 0.04;  $P = 0.900$ ). Smoking status ( $\beta = -0.02$ ; 95% CI: -0.05 to 0.01;  $P = 0.216$ ) was also not significantly associated with tortuosity (Figure 4, Supplemental Table 8). Using the classical tortuosity index (arc length/chord length), only age and RCA stenosis remained significant, and RCA stenosis showed a weak inverse association ( $\beta = -0.03$ ; 95% CI: -0.06 to 0.00;  $P = 0.042$ ) (Supplemental Table 10, Supplemental Figure 8). Our local tortuosity-based score showed modest agreement with the classical arc length/chord length index (Pearson  $r = 0.116$ ;  $P < 0.001$ ; Jaccard = 0.103 for top-decile overlap and 0.103 for bottom-decile overlap), and discordant examples demonstrate that our metric ranked visually tortuous vessels high while the classical index ranked them low, and vice versa (Supplemental Figure 9). We also tested whether tortuosity relates to the presence of multiple risk factors using a simple risk-factor count (hypertension, hypercholesterolemia, and smoking; 0-3), while modeling T2DM separately.

**FIGURE 3** Distribution of Right Coronary Artery Tortuosity by Sex and Age



(A) Difference in distribution of tortuosity for male and females. (B) Mean tortuosity by age groups of 5 years. Difference in mean tortuosity by demographics. A shows female patients have higher mean tortuosity than male patients, where the distribution of female patients is skewed to the right compared to men. Each line indicates the mean for each of the distributions. B shows that mean tortuosity increases with age across 5-year age groups. Sex-specific counts by 5-year age group used for B are shown in [Supplemental Table 13](#).

**FIGURE 4** Association of Right Coronary Artery Tortuosity With Cardiovascular Risk Factors and Presence of Stenosis in the Right Coronary Artery



Forest plot of adjusted associations with right coronary artery tortuosity per SD increase. Estimates are  $\beta$  coefficients from a multivariable linear regression model with age (standardized), sex (with female sex as reference), hypertension, type 2 diabetes mellitus, hypercholesterolemia, smoking, and presence of stenosis in the right coronary artery as covariates. Points show  $\beta$  with 95% CIs; the vertical line denotes  $\beta = 0$ . Tortuosity was higher in individuals with hypertension and presence of stenosis in the right coronary artery, but lower in people with diabetes. There was no association between tortuosity, hypercholesterolemia, and smoking. RCA = right coronary artery.

In an adjusted model that also accounted for CAD burden (Gensini score), the risk-factor count was not independently associated with tortuosity ( $\beta = 0.01$ ; 95% CI:  $[-0.01$  to  $0.03]$ ;  $P = 0.366$ ) (Supplemental Table 11).

**LOCAL TORTUOSITY DISCRIMINATES CAD BETTER THAN GLOBAL SCORE.** As shown in Supplemental Table 12, for predicting CAD in the RCA, a transformer neural network model having as inputs the full array of local tortuosity along with age and sex achieved an area under the receiver-operating characteristic curve of 0.67 (95% CI: 0.65-0.69), outperforming a logistic regression model based on mean tortuosity, age, and sex (area under the receiver-operating characteristic curve of 0.60; 95% CI: 0.58-0.63). These results indicate that leveraging the complete tortuosity profile provides greater discriminative power for detecting CAD than relying on the global score of mean curvature alone (Supplemental Figure 10).

## DISCUSSION

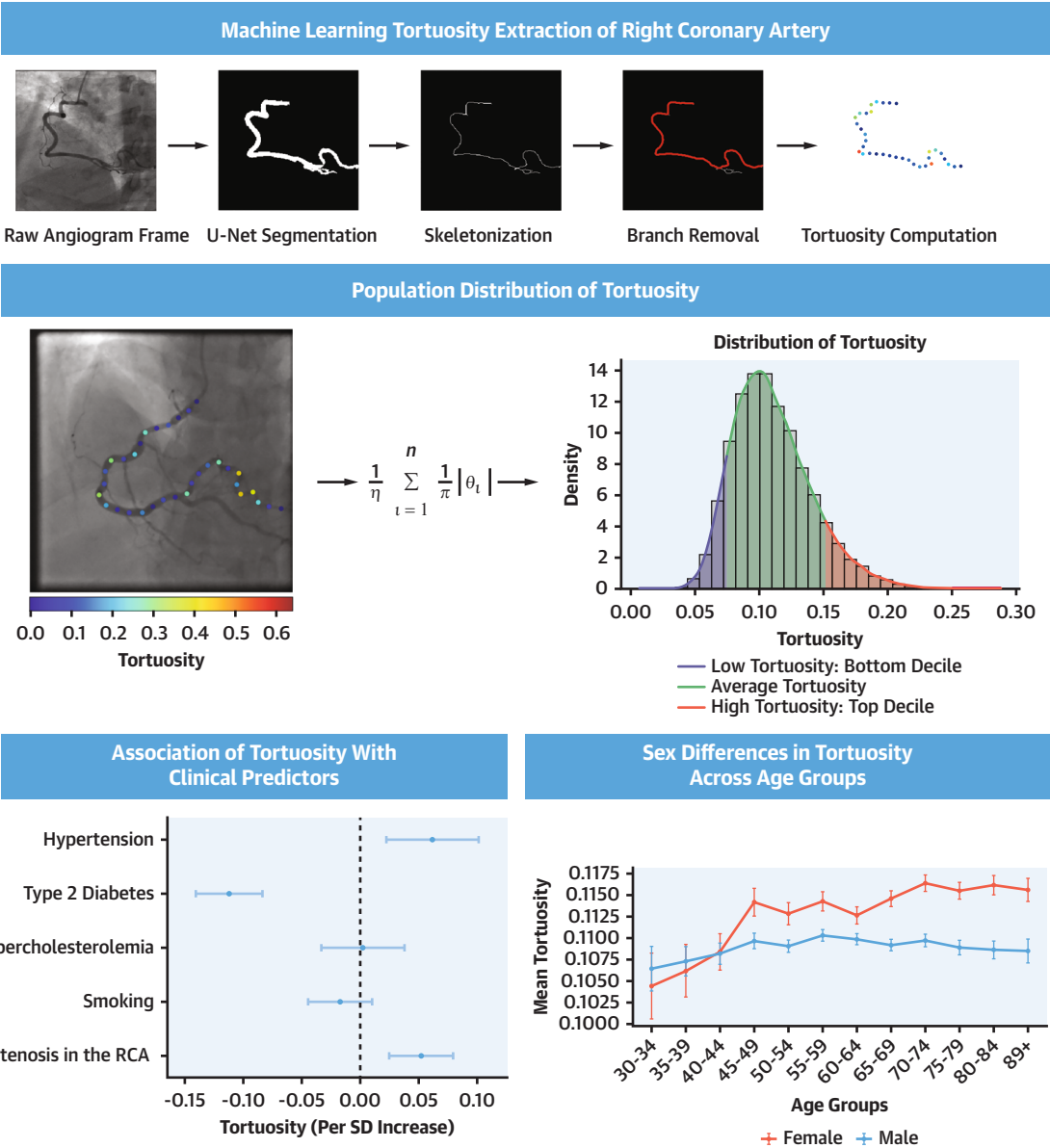
In this cohort of 38,691 coronary angiograms, we extracted a mathematically rigorous, scale- and rotation-invariant global tortuosity score and showed

that greater curvature is independently associated with CAD and hypertension but inversely related to T2DM. A fully automated deep-learning model that used local tortuosity features yielded a modest but significant gain in CAD discrimination, underscoring a potential biological link between arterial tortuosity and atherosclerosis (Central Illustration).

Tortuosity mirrored atherosclerotic burden. Patients with angiographically confirmed CAD exhibited higher curvature than those without CAD, contrasting with smaller studies that relied on heuristic measurements of tortuosity and reported inverse or unclear associations.<sup>18,19</sup> One plausible mechanism is that increased curvature produces regions of low or oscillatory wall shear stress, a well-established promoter of plaque formation.<sup>47</sup> Conversely, plaque deposition could stiffen the vessel wall and exaggerate curvature.<sup>48</sup> Regardless of directionality, our study demonstrates that tortuosity is more than an incidental angiographic finding.<sup>49</sup>

Coronary artery tortuosity may be clinically relevant through several mechanisms: 1) increased shear stress in tortuous vessel may accelerate atherosclerosis and plaque instability, as described above; 2) increased difficulty of delivering percutaneous interventions to nearby lesions and higher risk of stent-

**CENTRAL ILLUSTRATION** Machine Learning Extracted Tortuosity and Associations With Cardiovascular Risk, Coronary Artery Disease, and Sex Differences



Tello Ayala JR, et al. JACC Adv. 2026;5(6):102829.

(A) From raw angiographic frames, a U-Net segments the right coronary artery; the vessel is skeletonized, side branches are removed, and tortuosity is computed as the normalized sum of absolute turning angles along the centerline. (B) Population distribution of tortuosity in the study cohort, with “low” and “high” defined by the bottom and top deciles. (C) Associations of tortuosity with common clinical predictors shown as mean difference in tortuosity per SD (points) with 95% CIs: higher tortuosity with hypertension, lower with type 2 diabetes and angiographic right coronary artery stenosis, and no association with hypercholesterolemia and smoking. (D) Mean tortuosity across age groups stratified by sex, showing higher tortuosity in women compared with men from age 45 onward. Abbreviation as in [Figure 4](#).

related complications; and 3) serving as a marker of underlying arteriopathy. Prior studies of tortuosity have primarily focused on coronary angiograms, but computed tomography angiography-derived coronary tortuosity has also been associated with subclinical atherosclerosis and increased coronary artery calcium. The growing prevalence of computed tomography angiography provides an opportunity to noninvasively screen for tortuosity in larger cohorts.<sup>21,22,50-52</sup>

Traditional risk factors affected tortuosity in distinct ways. Hypertension, older age, and female sex each correlated positively with tortuosity, consistent with biomechanical and animal models showing that chronic pressure or reduced axial tension induces vessel elongation and kinking.<sup>18,20,53-55</sup> In women, hormonal differences and smaller arterial diameters may predispose to greater remodeling, while the mechanical advantage conferred before menopause likely reverses thereafter.<sup>24,25,56-59</sup> In a linear regression adjusted only for age and sex, older age showed a positive association with tortuosity; the correlation weakened when cardiovascular conditions were added to the model. As seen in **Figure 3B**, curvature increases qualitatively with age. Although these findings echo prior reports,<sup>22,23</sup> the attenuated effect after adjustment suggests that the observed increase in tortuosity with advancing age may be driven more by the cumulative development of cardiovascular pathology<sup>60</sup> than by age itself. This attenuation suggests that the apparent age effect may largely capture cumulative exposure to cardiovascular risk (eg, longer-standing hypertension) and its downstream vascular remodeling. We could not directly account for duration of risk factors or time on treatment, which may influence the degree of remodeling, but these time-dependent exposures are a plausible explanation for the loss of an independent age association after adjustment. In addition, women showed higher tortuosity from midlife onward, which may reflect time-dependent factors we could not measure, such as menopausal status. With respect to the inverse association between tortuosity and T2DM, similar observations have been made in retinal vessels,<sup>61</sup> where decreased tortuosity has been associated with an increased risk of ischemic heart disease. Hypercholesterolemia did not emerge as an independent determinant once other factors were considered, aligning with prior retinal observations.<sup>62</sup>

These findings position RCA tortuosity as an imaging biomarker that can be used in future studies to understand relationship of anatomy to cardiovascular risk factors and atherosclerosis formation

including better understanding of directionality and causality.<sup>22,25</sup> Tortuosity adds incremental information to conventional risk models—particularly in individuals with severe CAD.

**STUDY LIMITATIONS.** Our study focused on the RCA, anatomy which may not fully capture the more complex branching of the left coronary system<sup>21</sup> and thus may not be generalizable to the entire coronary tree. We focused on the RCA because its course and branching are more predictable and typically easier to segment reliably with an automated pipeline than the left coronary system, which has more complex bifurcations and more frequent overlap and foreshortening. In addition, the retrospective design can introduce selection bias and variability in imaging protocols. Segmentation was successful in 78.7% ( $n = 38,691$  of 49,189 eligible LAO RCA loops); the remainder were excluded during quality control, largely because real-world angiograms were not acquired with tortuosity measurement in mind (eg, suboptimal frame rate or limited orthogonal views) (**Supplemental Figure 1**). Future implementation-focused software will likely require minimum acquisition standards to reduce these exclusions as is the case of other angiography based physiology software currently in use.<sup>63</sup> Extension of this framework to the left coronary system will likely require analyzing the left anterior descending and left circumflex arteries separately rather than grouping the entire left tree into a single vessel, using standardized projections (eg, cranial views for left anterior descending-dominant visualization and caudal views for left circumflex-dominant visualization, when available). The segmentation component would require additional left circulation system labels, while the transformer model could use transfer learning from the current pipeline (eg, initializing with pretrained weights and fine-tuning on left-sided inputs) to support a unified model across vessels. Prospective validation and broader application of the discrete curvature framework—especially in evaluating the left coronary tree—will be necessary to confirm the utility of this metric in different populations and clinical settings. Nonetheless, our results lay the groundwork for future investigations, including prospective outcome studies that clarify how tortuosity interplays with other cardiovascular conditions and influences long-term cardiovascular events.

## CONCLUSIONS

We derived an automated and scalable measure of RCA tortuosity. Using a large data set, we demonstrate that tortuosity was higher in females, people

with hypertension and CAD, but lower in people with type 2 diabetes. Tortuosity is associated with CAD after adjusting for age, sex and risk factors, suggesting a potential role of this imaging biomarker in atherosclerosis pathophysiology. Overall, this study lays the groundwork for integrating ML approaches to assist the quantitative assessment of coronary artery morphology into our understanding of coronary atherosclerosis.

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## REFERENCES

- Martin SS, Aday AW, Allen NB, et al. 2025 Heart Disease and Stroke Statistics: a report of US and global data from the American Heart Association. *Circulation*. 2025;151(8):e41-e660.
- Ioannidis JPA, Trikalinos TA, Khoury MJ. Implications of small effect sizes of individual genetic variants on the design and interpretation of genetic association studies of complex diseases. *Am J Epidemiol*. 2006;164(7):609-614.
- Lopez Rodriguez M, Arasu UT, Kaikkonen MU. Exploring the genetic basis of coronary artery disease using functional genomics. *Atherosclerosis*. 2023;374:87-98.
- Gibbons RJ, Hodge D. The limitations of small outcome studies. *J Nucl Cardiol*. 2012;19(2):230-232.
- Enkiri S, Taylor A, Keeley E, Lipson L, Gimple L, Ragosta M. Coronary angiography is a better predictor of mortality than noninvasive testing in patients evaluated for renal transplantation. *Catheter Cardiovasc Interv*. 2010;76:795-801.
- Ciurica S, Lopez-Sublet M, Loeys BL, et al. Arterial tortuosity. *Hypertension*. 2019;73(5):951960.
- Weibel J, Fields WS. Tortuosity, coiling, and kinking of the internal carotid artery. II. Relationship of morphological variation to cerebrovascular insufficiency. *Neurology*. 1965;15:462-468.
- Lotmar W, Freiburghaus A, Bracher D. Measurement of vessel tortuosity on fundus photographs. *Albrecht Von Graefes Arch Klin Exp Ophthalmol*. 1979;211(1):49-57.
- Metz H, Bannister R, Murray-Leslie R, Bull J, Marshall J. Kinking of the internal carotid artery: in Relation to Cerebrovascular Disease. *Lancet*. 1961;277(7174):424-426.
- Ramos L, Novo J, Rouco J, Romeo S, Alvarez MD, Ortega M. Computational assessment of the retinal vascular tortuosity integrating domain-related information. *Sci Rep*. 2019;9(1):19940.
- Bullitt E, Gerig G, Pizer SM, Lin W, Aylward SR. Measuring tortuosity of the intracerebral vasculature from MRA images. *IEEE Trans Med Imaging*. 2003;22(9):1163-1171.
- Smedby O, Hogman N, Nilsson S, Erikson U, Olsson AG, Walldius G. Two-dimensional tortuosity of the superficial femoral artery in early atherosclerosis. *J Vasc Res*. 1993;30(4):181-191.
- Mateos MJ, Bribiesca E, Guzman-Arenas A, Aguilar W, Marquez-Flores JA. 3D tortuosity' computation as a shape descriptor and its application to brain structure analysis. *BMC Med Imaging*. 2024;24:130.
- Khurshid S, Friedman S, Reeder C, et al. ECG-based deep learning and clinical risk factors to predict atrial fibrillation. *Circulation*. 2022;145(2):122-133.
- Pirruccello JP, Chaffin MD, Chou EL, et al. Deep learning enables genetic analysis of the human thoracic aorta. *Nat Genet*. 2022;54(1):40-51.
- Agrawal S, Klarqvist MDR, Diamant N, et al. BMI-adjusted adipose tissue volumes exhibit depot-specific and divergent associations with cardiometabolic diseases. *Nat Commun*. 2023;14(1):266.
- Haas ME, Pirruccello JP, Friedman SN, et al. Machine learning enables new insights into genetic contributions to liver fat accumulation. *Cell Genom*. 2021;1(3):100066.
- Li Y, Shen C, Ji Y, Feng Y, Ma G, Liu N. Clinical implication of coronary tortuosity in patients with coronary artery disease. *PLoS One*. 2011;6(8):e24232.
- Khosravani-Rudpishi M, Joharimoghadam A, Rayzan E. The significant coronary tortuosity and atherosclerotic coronary artery disease; what is the relation? *J Cardiovasc Thorac Res*. 2018;10(4):209-213.
- Pancera P, Ribul M, Presciutti B, Lechi A. Prevalence of carotid artery kinking in 590 consecutive subjects evaluated by Echocolor Doppler. Is there a correlation with arterial hypertension? *J Int Med*. 2000;248(1):7-12.
- Eleidi MF, Guddeti RR, Tweet MS, et al. Coronary artery tortuosity in spontaneous coronary artery dissection. *Circulation*. 2014;7(5):656-662.
- Zebic Mihic P, Saric S, Bilic Curcic I, Mihaljevic I, Juric I. The association of severe coronary tortuosity and non-obstructive coronary artery disease. *Medicina (Kaunas)*. 2023;59(9):1619.
- Li M, Wang ZW, Fang LJ, Cheng SQ, Wang X, Liu NF. Correlation analysis of coronary artery tortuosity and calcification score. *BMC Surg*. 2022;22(1):66.
- Zaid M, Sala L, Despina L, et al. Cardiovascular sex-differences: insights via physiology-based modeling and potential for noninvasive sensing via ballistocardiography. *Front Cardiovasc Med*. 2023;10:1215958.
- Burgess SN. Understudied, under-recognized, underdiagnosed, and undertreated: sex based disparities in cardiovascular medicine. *Circulation*. 2022;15(2):e011714.
- Yang S, Kweon J, Roh JH, et al. Deep learning segmentation of major vessels in X-ray coronary angiography. *Sci Rep*. 2019;9(1):16897.
- Kaba S, Hacı H, Isin A, İlhan A, Konkbayır C. The application of deep learning for the segmentation and classification of coronary arteries. *Diagnostics (Basel)*. 2023;13(13):2274.
- Popov M, Amanturdieva A, Zhaksylyk N, et al. Dataset for automatic region-based coronary artery disease diagnostics using X-Ray angiography images. *Sci Data*. 2024;11(1):20.

29. Rigatelli G, Gianese F, Zuin M. Modern atlas of invasive coronary angiography views: a practical approach for fellows and young intervention-alists. *Int J Cardiovasc Imaging*. 2022;38(5):919-926.
30. Kuon E, Dahm JB, Empen K, Robinson DM, Reuter G, Wucherer M. Identification of less irra-diating tube angulations in invasive cardiology. *J Am Coll Cardiol*. 2004;44(7):1420-1428.
31. Ronneberger O, Fischer P, Brox T. U-Net: Convolutional networks for biomedical image segmentation. In: Navab N, Hornegger J, Wells WM, Frangi AF, eds. *Medical image computing and computer-assisted intervention – MICCAI 2015*. Cham: Springer International Pub-lishing; 2015.
32. Sianos G, Morel MA, Kappetein AP, et al. The SYNTAX score: an angiographic tool grading the complexity of coronary artery disease. *Euro-Intervention*. 2005;1(2):219-227.
33. Dice LR. Measures of the amount of ecologic association between species. *Ecology*. 1945;26(3): 297-302.
34. Zijdenbos A, Dawant B, Margolin R, Palmer A. Morphometric analysis of white matter lesions in MR images: method and validation. *IEEE Trans Med Imaging*. 1994;13(4):716-724.
35. Zou KH, Warfield SK, Bharatha A, et al. Sta-tistical validation of image segmentation quality based on a spatial overlap index. *Acad Radiol*. 2004;11(2):178-189.
36. Gould K, Lipscomb K, Hamilton GW. Physio-logic basis for assessing critical coronary stenosis: instantaneous flow response and regional distri-bution during coronary hyperemia as measures of coronary flow reserve. *Am J Cardiol*. 1974;33(1): 87-94.
37. Crane K, Wardetzky M. A glimpse into discrete differential geometry. *Notices Am Math Soc*. 2017;64:1153-1159.
38. Bribiesca E. A measure of tortuosity based on chain coding. *Pattern Recognition*. 2013;46(3): 716-724.
39. Kashyap V, Gharlegghi R, Li DD, et al. Accuracy of vascular tortuosity measures using computa-tional modelling. *Sci Rep*. 2022;12(1):865.
40. Shivaie S, Tohidi H, Loganathan P, Kar M, Hashemy H, Shafiee MA. Interobserver variability of coronary stenosis characterized by coronary angiography: a single-center (Toronto General Hospital) retrospective chart review by staff car-diologists. *Vasc Health Risk Manag*. 2024;20:359-368.
41. Rampidis GP, Benetos G, Benz DC, Giannopoulos AA, Buechel RR. A guide for Gensini Score calculation. *Atherosclerosis*. 2019;287:181-183.
42. Wiley LK, Shah A, Xu H, Bush WS. ICD-9 to-bacco use codes are effective identifiers of smoking status. *J Am Med Inform Assoc*. 2013;20(4):652658.
43. Shetty NS, Gaonkar M, Patel N, et al. Trends of lipid concentrations, awareness, evaluation, and treatment in severe dyslipidemia in US adults. *Mayo Clin Proc*. 2024;99(2):271-282.
44. Benn M, Watts G, Tybjaerg-Hansen A, Nordestgaard B. Familial hypercholesterolemia in the Danish general population: prevalence, coro-nary artery disease, and cholesterol-lowering medication. *J Clin Endocrinol Metab*. 2012;97: 3956-3964.
45. Law MR, Wald NJ, Morris JK, Jordan RE. Value of low dose combination treatment with blood pressure lowering drugs: analysis of 354 rando-mised trials. *BMJ*. 2003;326(7404):1427.
46. Balakrishnan P, Beaty TH, Young JH, Colantuoni E, Matsushita K. Methods to estimate underlying blood pressure: the Atherosclerosis Risk in Communities (ARIC) study. *PLoS One*. 2017;12:e0179234.
47. Malek AM, Alper SL, Izumo S. Hemodynamic shear stress and its role in atherosclerosis. *JAMA*. 1999;282(21):2035-2042.
48. Han HC. Twisted blood vessels: symptoms, etiology and biomechanical mechanisms. *J Vasc Res*. 2012;49(3):185-197.
49. Hathout L, Do HM. Vascular tortuosity: a mathematical modeling perspective. *J Physiol Sci*. 2012;62:133-145.
50. Kahe F, Sharfaei S, Pitliya A, et al. Coronary artery tortuosity: a narrative review. *Coronary Artery Dis*. 2020;31(2):187-192.
51. Konigstein M, Ben-Yehuda O, Redfors B, et al. Impact of coronary artery tortuosity on outcomes following stenting. *JACC Cardiovasc Interv*. 2021;14(9):1009-1018.
52. El Tahlawi M, Sakrana A, Elmurr A, Gouda M, Tharwat M. The relation between coronary tor-tuosity and calcium score in patients with chronic stable angina and normal coronaries by CT angi-ography. *Atherosclerosis*. 2016;246:334-337.
53. Jakob M, Spasojevic D, Krogmann N, Wiher H, Hug R, Hess M. Tortuosity of coronary arteries in chronic pressure and volume overload. *Cathet Cardiovasc Diagn*. 1996;38(1):25-31.
54. Han HC. A biomechanical model of artery buckling. *J Biomech*. 2007;40(16):3672-3678.
55. Sho E, Nanjo H, Sho M, et al. Arterial enlarge-ment, tortuosity, and intimal thickening in response to sequential exposure to high and low wall shear stress. *J Vasc Surg*. 2004;39(3):601-612.
56. Natoli AK, Medley TL, Ahimastos AA, et al. Sex steroids modulate human aortic smooth muscle cell matrix protein deposition and matrix metalloproteinase expression. *Hypertension*. 2005;46(5):1129-1134.
57. Wentzel JJ, Papafakis MI, Antoniadis AP, et al. Sex-related differences in plaque charac-teristics and endothelial shear stress related plaque-progression in human coronary arteries. *Atherosclerosis*. 2022;342:9-18.
58. Moreau KL, Babcock MC, Hildreth KL. Sex differences in vascular aging in response to testosterone. *Biol Sex Differ*. 2020;11(1):18.
59. Seeland U, Nemcsik J, Lønnebakken MT, et al. Sex and gender aspects in vascular ageing – focus on epidemiology, pathophysiology, and out-comes. *Heart Lung Circ*. 2021;30(11):1637-1646.
60. Kannel WB, Vasan RS. Is age really a non-modifiable cardiovascular risk factor? *Am J Car-diol*. 2009;104(9):1307-1310.
61. Owen CG, Newsom RS, Rudnicka AR, Barman SA, Woodward EG, Ellis TJ. Diabetes and the tortuosity of vessels of the bulbar conjunc-tiva. *Ophthalmology*. 2008;115(6):e27-e32.
62. Klein R, Klein BE, Tomany SC, Cruickshanks KJ. The association of cardiovascular disease with the long-term incidence of age-related maculopathy: the beaver dam eye study. *Ophthalmology*. 2003;110(6):1273-1280.
63. Pellicano M, Lavi I, Bruyne BD, et al. Validat-ion study of image-based fractional flow reserve during coronary angiography. *Circulation*. 2017;135(9):e005259.

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**KEY WORDS** angiography, artificial intelligence, coronary artery disease, tortuosity

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**APPENDIX** For supplemental tables and figures, please see the online version of this paper.